Unit-3

Baseband Transmission

→ comes under short dist. comm.
→ No need for modulation
→ Msg is sent as it is through the channel.

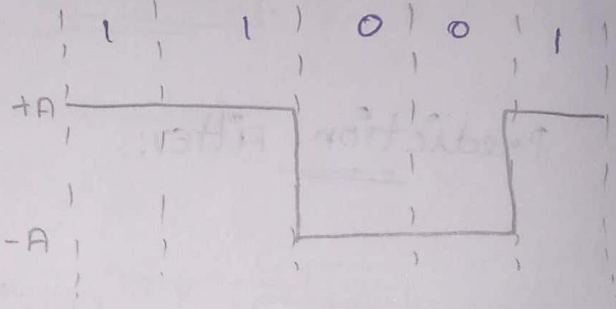
we are not sending as it is instead we send pulses

Line coding method → converting digital data into pulses.

Line coding - process of converting binary data into a digital signal. There digital signals must be compatible with the characteristics of baseband channel.

Eg: Binary data 1 1 0 0 1

Digital signal



Properties of Line coding:

1) Signal level and data level
2) Pulse rate and bit rate
3) Dc component
4) Self-synchroniyation
5) Transmission BW
6) Low probability of error
7) Transmission power efficiency
8) Noise and interference tolerance

signal level and data level:

signal level:

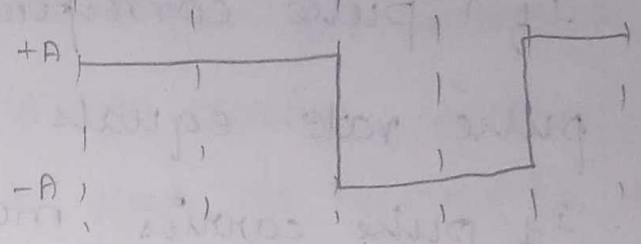
The no. of values allowed to represent a particular signal is defined as no. of signal levels.

Data level:

The no. of values used to represent the data is called no. of data levels.

Eg-1: Binary data 1 1 1 0 0 1

Digital signal



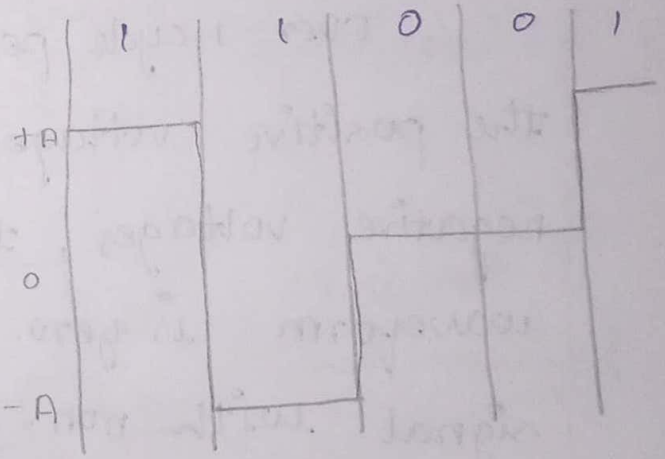
Data level = 2 (0, 1)

signal level = 2 (+A, -A)

Eg-2:

Binary data 1 1 0 0 1

Digital signal



initial => +A

successive / consecutive 1's => invert the polarity

0 => zero level

Data level = 2 (0, 1)

signal level = 3 (A, 0, -A)

In both examples, pulse rate = bit rate

2) Pulse rate and Bit rate:

Pulse rate:- is defined as no. of pulses transmitted per second. A pulse is defined as min. amount of time required to transmit a symbol.

Bit rate :- is defined as no. of bits transmitted per second.

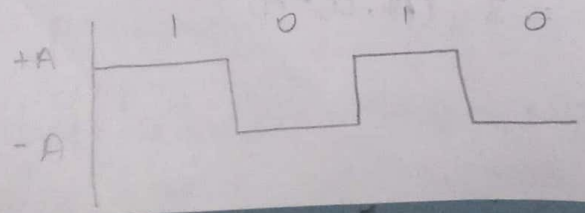
If 1 pulse corresponds to 1 bit, then pulse rate equals the bit rate.

If pulse carries more than 1 bit, then pulse rate is lower than the bit rate.

3) DC component:

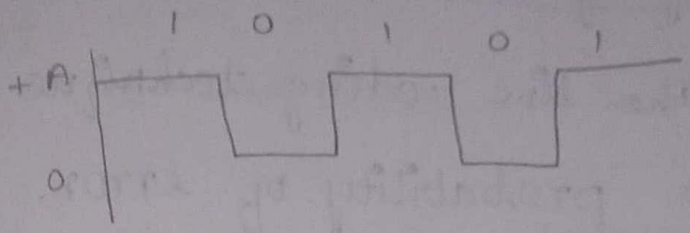
Over 1 cycle period of waveform, if all the positive voltages are cancelled by negative voltages, then DC component of the waveform is zero. In line coding, the signal with non-zero dc component is treated as distorted one and it creates error in received signal.

Eg: 1 → +A
0 → -A



1 → +A

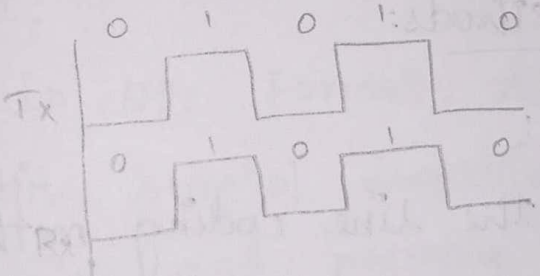
0 → 0



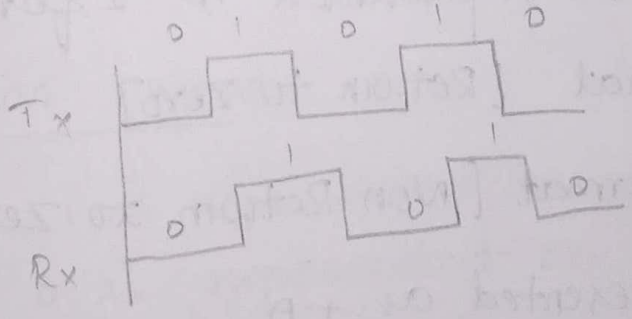
Average value = +A

4) Self - synchronization:

If the receiver bit intervals corresponds exactly to the transmitter bit intervals, then it is possible to receive signals correctly. If clock freq. at Tx and Rx are different then the data at the Tx and Rx is different resulting in error.



clock is synchronized



clk is not synchronized

5) Transmission BW:

The line coding technique should have lesser bandwidth.

6) Probability of error:

The line coding technique should have low probability of error.

7) Transmission power efficiency:

The transmission power should have low power requirement for line coding technique.

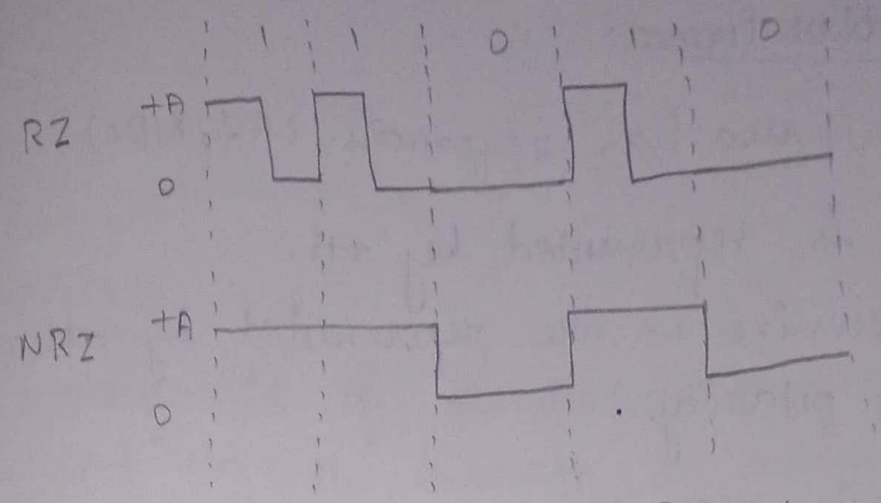
8) Noise and interference tolerance:

The line coding method should have high noise tolerance.

Line Coding Methods:

1. Unipolar format:

→ It is one of the line coding method.
→ This has is represented in 2 formats
  => RZ format [Return to Zero]
  => NRZ format [Non Return to Zero]
=> 1 is represented as +A,
  0 is represented as 0.



RZ => after half pulse, it will return to zero.
(or at middle)

RI:

In RZ format, the pulses occupy a fraction of symbol duration i.e., the signal returns to zero at the middle of pulse duration.

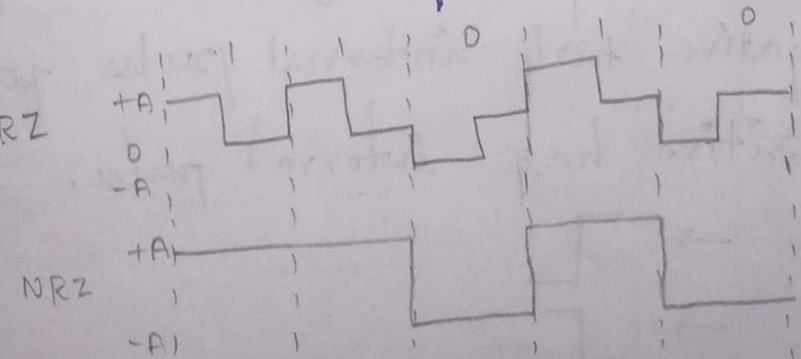
NRI:

In NRZ Format, the pulses occupy entire symbol duration. It remains at +A for '1' and returns to zero for '0'.

2. Polar Format:

=> 1 is represented as +A
0 is represented as -A

=> It has two formats (RZ, NRZ).



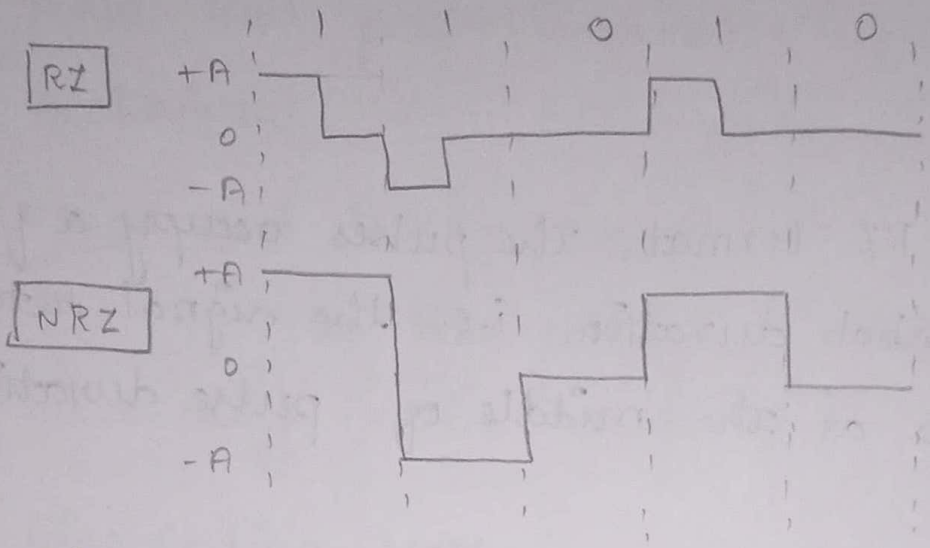
3. Bipolar Format:

=> It also has 2 formats. (RZ, NRZ).

=> 1 is represented by +A.

=> Successive 1's are represented by alternating the polarity.

=> 0 is rep. as 0.



Advantages:

=> If there are even no. of 1's then the DC component of bipolar format is 0.

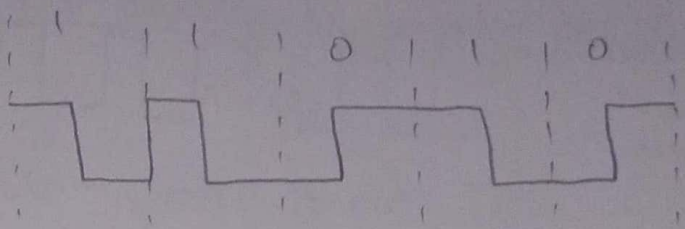
Split Phase Manchester:

=> It is a line coding technique.

=> Here, symbol 1 is represented by positive half interval pulse followed by negative half interval pulse. Symbol 0 is rep. by negative half interval pulse followed by positive half interval pulse.

1 → ⌐¬

0 → ¬⌐



Advantage:

⇒ Irrespective of the probability of occurence of 0's and 1's, the average Dc component is zero. Hence power savings is more in this technique.

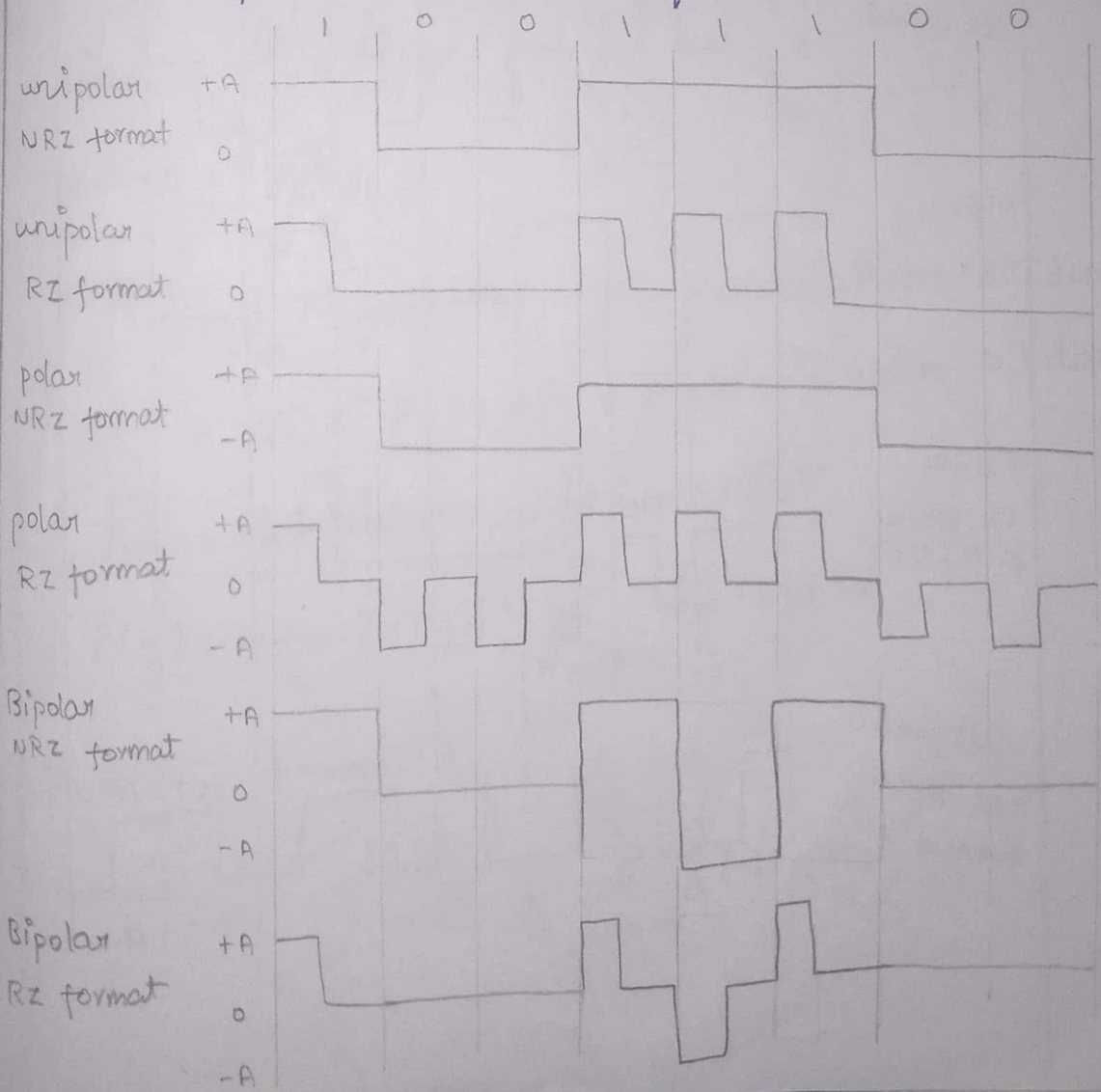
1. Represent the data using i) unipolar RZ, NRZ format

A) 1 0 0 1 1 1 0 0 B) 1 1 1 0 0 1 0 1

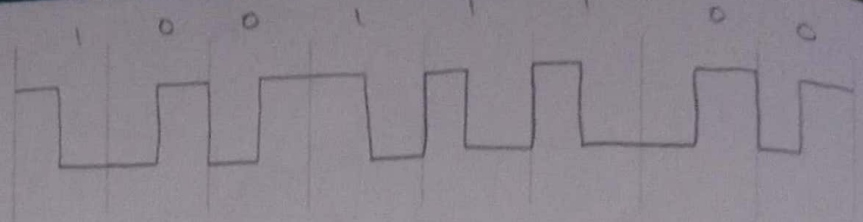
ii) Polar RZ, NRZ format iii) Bipolar RZ, NRZ Format

iv) split phase manchester format.

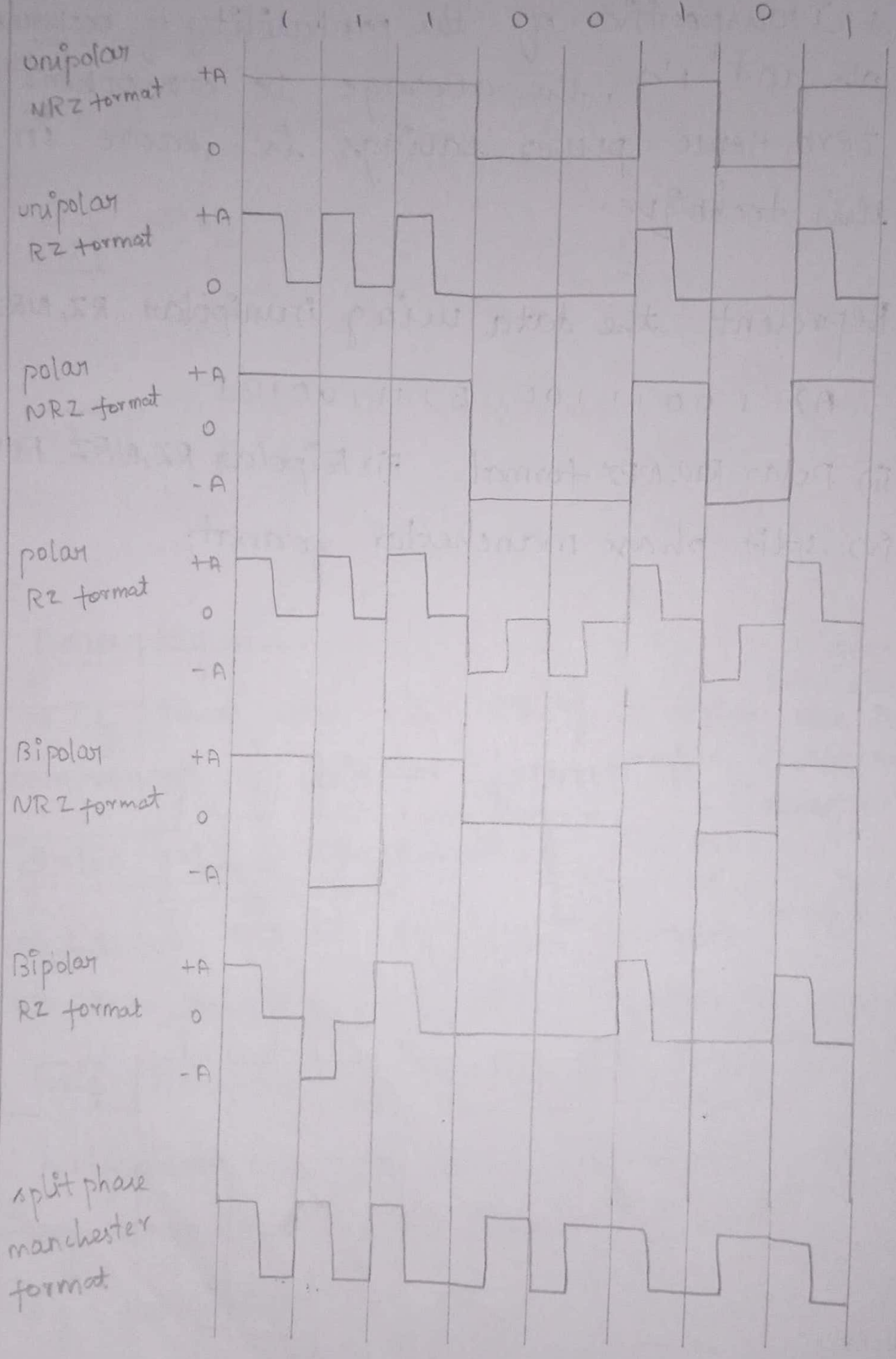
A)



split phase manchester format



B)



Power spectral Density of unipolar NRZ Line Coding scheme:

A power spectral density [PSD] is a measure of signal power content vs frequency. PSD is typically used to characterize a broad band random signal.

Auto-correlation - represents the degree of similarity b/t a given time series and the lagged version of itself over successive time intervals.

Note:

1. $X(f) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$

2. $\sin\theta = e^{j\theta} - e^{-j\theta} / 2$

3. $\text{sinc}\theta = \frac{\sin\theta}{\theta}$

4. $E(X^2) = \int_{-\infty}^{\infty} x^2 f_x(x) dx$ → Autocorrelation C.T domain

5. $E(X^2) = \sum x^2 P(X = x)$ → Autocorrelation D.T domain

6. wiener khintchin's relation (PSD):

$$P(f) = \frac{1}{T_b} |X(f)|^2 \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

7. Poisson Formula:

$$\sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_b} = \frac{1}{T_b} \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right)$$

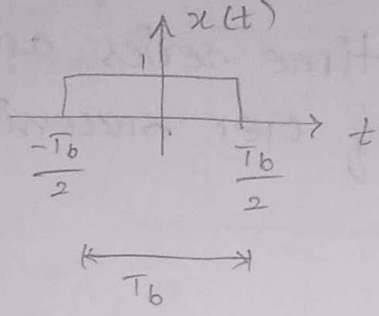
Step 1: Find FT $X(f)$ of NRZ pulse.

Step 2: Find auto-correlation $R_A(n)$ of NRZ pulse

Step 3: Calculate PSD based on $X(f)$ and $R_A(n)$

PSD of unipolar NRZ pulse:

1. Find FT $X(f)$ of unipolar NRZ pulse.



$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$= \int_{\frac{-T_b}{2}}^{T_b/2} (1) e^{-j2\pi ft} dt$$

$$= \left[\frac{e^{-j2\pi ft}}{-j2\pi f} \right]_{-T_b/2}^{T_b/2}$$

$$= \frac{e^{\frac{-j2\pi fT_b}{2}} - e^{\frac{j2\pi fT_b}{2}}}{-j2\pi f}$$

$$= \frac{-1}{-\pi f} \left[\frac{e^{\frac{j2\pi fT_b}{2}} - e^{\frac{-j2\pi fT_b}{2}}}{2j} \right]$$

$$= \frac{1}{\pi f} \sin \frac{2\pi fT_b}{2}$$

$$= \frac{1}{\pi f} \sin(\pi f T_b)$$

$$= \frac{T_b}{T_b} \left[\frac{\sin(\pi f T_b)}{\pi f} \right]$$

$$X(f) = T_b \operatorname{sinc}(\pi f T_b)$$

2. Find auto-correlation $R_A(n)$ of unipolar NRZ pulse.

=> Assume the probability occurrence of 0's and 1's is equal.

$$P(A_L = 1) = P(A_L = 0) = \frac{1}{2}$$

=> unipolar : $1 \rightarrow +A$

$0 \rightarrow 0$

$$R_A[n] = E[A_L A_{L-n}]$$

i) $n = 0$:

$$R_A[0] = E[A_L A_L]$$

Bit	A_L	A_L	A_L^2	Prob
0	0	0	0	1/2
1	+A	+A	A^2	1/2

$$E[X^2] = \sum x^2 P(X = x)$$

$$R_A[0] = \sum A_L^2 P(X = x)$$

$$= 0\left(\frac{1}{2}\right) + A^2\left(\frac{1}{2}\right)$$

$$R_A[0] = \frac{A^2}{2}$$

ii) $n \neq 0$:

$$R_A(n) = E[A_L . A_{L-n}]$$

Bit	A_L	A_{L-n}	$A_L . A_{L-n}$	Prob
0 0	0	0	0	1/4
0 1	0	+A	0	1/4
1 0	+A	0	0	1/4
1 1	+A	+A	A^2	1/4

$$R_A[n] = 0\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) + A^2\left(\frac{1}{4}\right) = \frac{A^2}{4}$$

$$\therefore R_A(n) = \begin{cases} \frac{A^2}{2}, & n = 0 \\ \frac{A^2}{4}, & n \neq 0 \end{cases}$$

3. Calculate PSD of unipolar NRZ based on $X(f)$ and $R_A(n)$.

$$P(f) = \frac{1}{T_b} |X(f)|^2 \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

$$= \frac{1}{T_b} \left| T_b \operatorname{sinc}(\pi f T_b) \right|^2 \left[\frac{A^2}{2}(1) + \frac{A^2}{4} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right]$$

$$= \frac{1}{T_b} T_b^2 \operatorname{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} + \frac{A^2}{4} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right]$$

$$= T_b \operatorname{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} + \frac{A^2}{4} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right]$$

$$= \frac{A^2}{2} T_b \operatorname{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \operatorname{sinc}^2(\pi f T_b) \left[\left\{ \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right\} + 1 - 1 \right]$$

$$= \frac{A^2}{2} T_b \operatorname{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \operatorname{sinc}^2(\pi f T_b) \left[\left\{ \sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_b} \right\} - 1 \right]$$

$$P(f) = \frac{A^2}{2} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b)(-1)$$

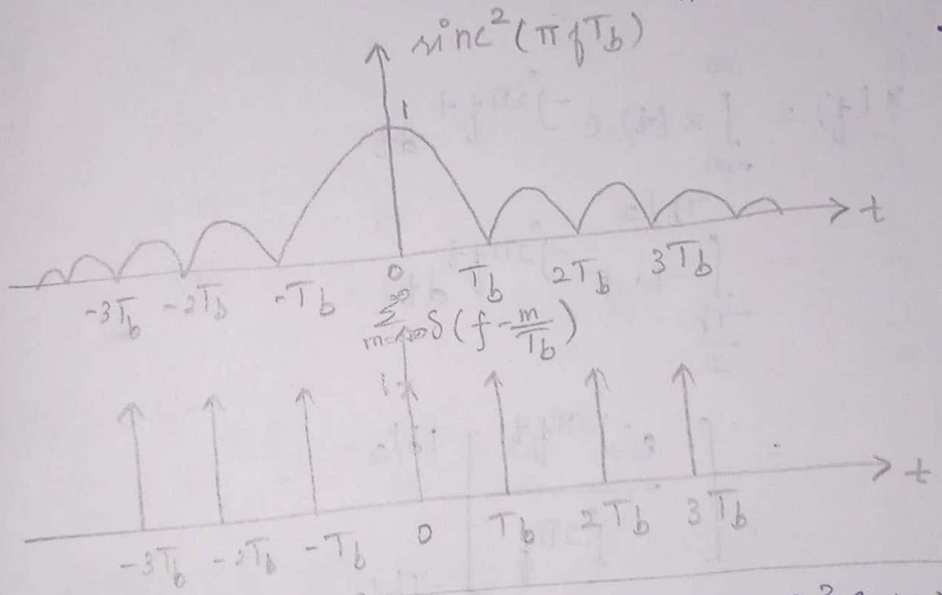
$$+ \frac{A^2}{4}\left[\sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_b}\right] (T_b \text{sinc}^2(\pi f T_b))$$

Apply Poisson's formula

$$= \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b)$$

$$\left[\frac{1}{T_b}\sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right)\right]$$

$$P(f) = \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} \text{sinc}^2(\pi f T_b) \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right) \rightarrow (1)$$



(1) $\Rightarrow$ $\boxed{P(f) = \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4}\delta(f)}$ $\rightarrow$ (2)

$\frac{A^2}{4}\delta(f)$: DC component.

Eqn. (2) represents the PSD of unipolar NRZ format. The second term $\frac{A^2}{4}\delta(f)$ is the DC component present in the unipolar NRZ format.

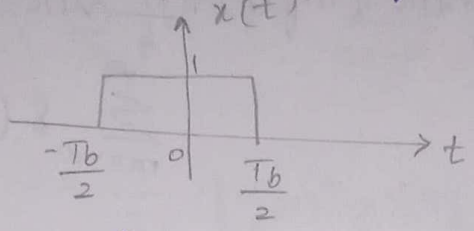
PSD of NRZ Polar format:

Step 1: Find FT $X(f)$ of NRZ polar pulse.

Step 2: Find autocorrelation $R_A(n)$ of NRZ polar pulse.

Step 3: Calculate PSD based on $X(f)$ and $R_A(n)$.

Step 1: Find the FT $X(f)$ of NRZ polar pulse.



$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$= \int_{\frac{-Tb}{2}}^{Tb/2} 1 \cdot e^{-j2\pi ft}\, dt$$

$$= \left[\frac{e^{-j2\pi ft}}{j2\pi f} \right]_{-Tb/2}^{Tb/2}$$

$$X(f) = T_b \operatorname{sinc}(\pi f T_b)$$

Step 2: Autocorrelation $R_A(n) = E[A_L \cdot A_{L-n}]$

i) $n = 0$

$$R_A(0) = E[A_L\, A_L]$$

Bit	A_L	A_L	A_L^2	Probability
0	$-A$	$-A$	A^2	$1/2$
1	$+A$	$+A$	A^2	$1/2$

General eqn:

$$E[x^2] = \sum x^2 P(X=x)$$

$$R_A(0) = \sum x^2 P(X=x)$$

$$= A^2\left(\frac{1}{2}\right) + A^2\left(\frac{1}{2}\right)$$

$$R_A(0) = A^2$$

ii) $n \neq 0$

$$R_A(n) = E[A_L \; A_{L-n}]$$

Bit	A_L	A_{L-n}	$A_L . A_{L-n}$	Probability
0 0	$-A$	$-A$	A^2	$\frac{1}{4}$
0 1	$-A$	$+A$	$-A^2$	$\frac{1}{4}$
1 0	$+A$	$-A$	$-A^2$	$\frac{1}{4}$
1 1	$+A$	$+A$	A^2	$\frac{1}{4}$

$$R_A(n) = \frac{A^2}{4} - \frac{A^2}{4} - \frac{A^2}{4} + \frac{A^2}{4} = 0$$

$$R_A(n) = \begin{cases} A^2, & n=0 \\ 0, & n \neq 0 \end{cases}$$

Step 3: calculate PSD of polar NRZ pulse.

$$P(f) = \frac{1}{T_b} |X(f)|^2 \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

$$= \frac{1}{T_b} \left| T_b \operatorname{sinc}(\pi f T_b) \right|^2 [A^2 + 0]$$

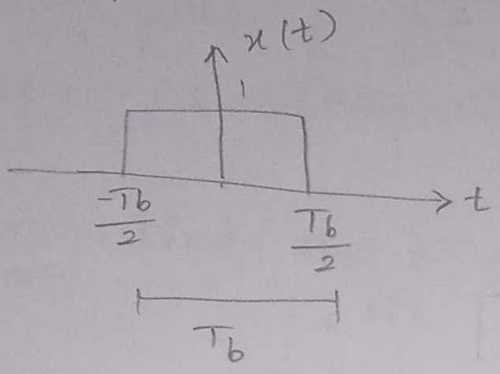
$$P(f) = T_b \operatorname{sinc}^2(\pi f T_b) A^2 \longrightarrow (1)$$

Conclusion: No DC component is present in

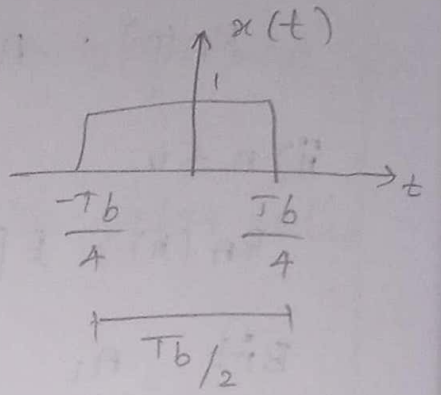
① Hence it is better than the NRZ unipolar format.

Note:

NRZ pulse.



RZ pulse.



Step 1: Change the pulse duration from (-Tb/2 to Tb/2) to (-Tb/4 to Tb/4)

Step 2: and step 3 remains same.

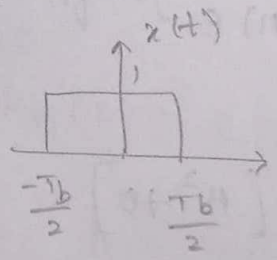
PSD of NRZ Bipolar Format:

Step 1: Find FT X(f) of NRZ Bipolar pulse.

Step 2: Find autocorrelation $R_A(n)$ of NRZ Bipolar pulse.

Step 3: Calculate PSD based on X(f) and $R_A(n)$.

Step 1:



same as unipolar NRZ pulse.

$$X(f) = \int_{-Tb/2}^{Tb/2} x(t) e^{-j2\pi f ct} dt$$

$$x(f) = \int_{\frac{-T_b}{2}}^{T_b/2} 1 \cdot e^{-j2\pi f_c t} \, dt$$

$$x(f) = T_b \operatorname{sinc}(\pi f T_b)$$

Step 2: Find autocorrelation $R_A(n)$.

Assume probability of occurence of 0's and 1's are equal.

Bipolar format: $0 \longrightarrow 0$

$1 \longrightarrow +A$

for consecutive 1's change polarity.

$$R_A(n) = E[A_L \; A_{L-n}]$$

A_{L-n} is shifted version of A_L.

i) $n = 0$:

$$R_A(0) = E[A_L \; A_L]$$

Bit	A_L	A_L	A_L^2	probability
0	0	0	0	1/2
1	A	A	A^2	1/2

$$R_A(0) = 0 + \frac{A^2}{2}$$

$$R_A(0) = \frac{A^2}{2}$$

ii) $n = \pm 1$

$$R_A(1) = E[A_L \; A_{L-1}]$$

Bit	A_L	A_{L-1}	$A_L A_{L-1}$	prob	
0 0	0	0	0	1/4	
0 1	0	$\pm A$	0	1/4	
1 0	$\pm A$	0	0	1/4	
1 1	$+A$	$-A$	$-A^2$	1/2	1/4
	$-A$	$+A$	$-A^2$	1/2	

$$R_A(1) = \frac{-A^2}{8} - \frac{A^2}{8} = \frac{-A^2}{4}$$

iii) $n \geq 2$

$$R_A(n) = E[A_L \cdot A_{L-n}].$$

Bit	A_L	A_{L-n}	$A_L A_{L-n}$	prob	
0 0	0	0	0	1/4	
0 1	0	$\pm A$	0	1/4	
1 0	$\pm A$	0	0	1/4	
1 1	$+A$	$-A$	$-A^2$	1/16	1/4
	$-A$	$+A$	$-A^2$	1/16	
	$+A$	$+A$	A^2	1/16	
	$-A$	$-A$	$+A^2$	1/16	

$$R_A(n) = \frac{-A^2}{16} - \frac{A^2}{16} + \frac{A^2}{16} + \frac{A^2}{16} = 0.$$

$$R_A(n) = \begin{cases} A^2/2, & n = 0 \\ -A^2/4, & n = \pm 1 \\ 0, & n \geq 2 \end{cases}$$

Step 3: $$P(f) = \frac{1}{T_b} |X(f)|^2 \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

$$P(f) = \frac{1}{T_b} \left| T_b \operatorname{sinc}(\pi f T_b) \right|^2 \left[\underbrace{\frac{A^2}{2} \cdot 1}_{n=0} + \underbrace{\left(-\frac{A^2}{4}\right) e^{-j2\pi f T_b}}_{n=+1} + \underbrace{\left(-\frac{A^2}{4}\right) e^{j2\pi f T_b}}_{n=-1} + \underbrace{0}_{n \geq 2} \right]$$

$$= T_b \operatorname{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} - \frac{A^2}{4} e^{-j2\pi f T_b} - \frac{A^2}{4} e^{j2\pi f T_b} \right]$$

$$= T_b \operatorname{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} - \frac{A^2}{4} \left(e^{j2\pi f T_b} + e^{-j2\pi f T_b} \right) \right]$$

$$= T_b \operatorname{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} - \frac{A^2}{4} (2\cos(2\pi f T_b) \right]$$

$$= \frac{A^2}{2} T_b \operatorname{sinc}^2(\pi f T_b) - \frac{A^2}{2} \cos(2\pi f T_b)$$

$$T_b \operatorname{sinc}^2 |\pi f T_b|$$

$$= \frac{A^2}{2} T_b \operatorname{sinc}^2(\pi f T_b) \left[1 - \cos(2\pi f T_b) \right]$$

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

$$P(f) = \frac{A^2}{2} T_b \operatorname{sinc}^2(\pi f T_b) \left(2 \sin^2(2\pi f T_b) \right)$$

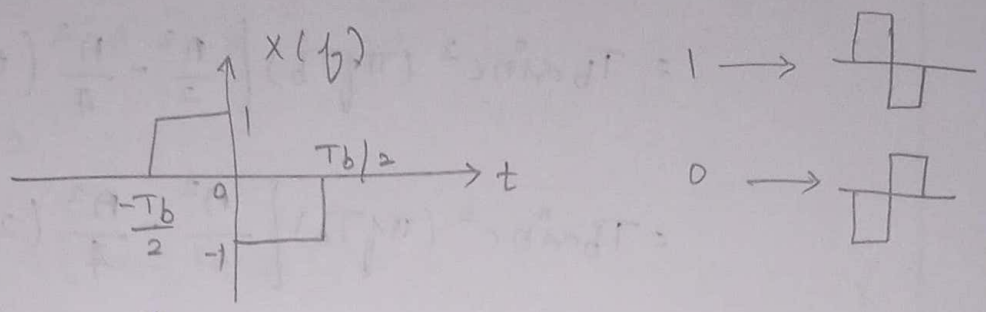
$$P(f) = A^2 T_b \operatorname{sinc}^2(\pi f T_b) \sin^2(\pi f T_b)$$

PSD of manchester polar format:

Step 1: Find FT of $x(t)$ of manchester polar format.

Step 2: Find autocorrelation $R_A(n)$.

Step 3: Find PSD $P(f)$ based on $x(f)$ and $R_A(n)$.



$$X(f) = \int_{-\infty}^{\infty} x(f)\, e^{-j2\pi ft}\, dt$$

$$= \left[\int_{\frac{-T_b}{2}}^{0} 1 \cdot e^{-j2\pi ft} + \int_{0}^{T_b/2} (-1) \cdot e^{-j2\pi ft} \right] dt$$

$$= \left[\frac{e^{-j2\pi ft}}{-j2\pi f} \right]_{-T_b/2}^{0} - \left[\frac{e^{-j2\pi ft}}{-j2\pi f} \right]_{0}^{T_b/2}$$

$$= \frac{-1}{j2\pi f} \left[1 - e^{-j2\pi f (-T_b/2)} \right] + \frac{1}{j2\pi f} \left[e^{-j2\pi f \frac{T_b}{2}} - 1 \right]$$

$$= \frac{-1}{j2\pi f} \left[1 - e^{j\pi f T_b} \right] + \frac{1}{j2\pi f} \left[e^{-j2\pi f T_b} - 1 \right]$$

$$= -\frac{1}{j2\pi f} + \frac{e^{j\pi f T_b}}{j2\pi f} + \frac{e^{-j\pi f T_b}}{j2\pi f} - \frac{1}{j2\pi f}$$

$$= \frac{-1}{j\pi f} + \frac{1}{j2\pi f}\left[e^{j\pi f T_b} + e^{-j\pi f T_b}\right]$$

$$= \frac{-1}{j\pi f} + \frac{1}{j2\pi f}\left[2\cos(\pi f T_b)\right]$$

$$X(f) = \frac{-1}{j\pi f} + \frac{\cos(\pi f T_b)}{j\pi f}$$

$$\because \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$X(f) = \frac{-1}{j\pi f}\left[1-\cos(\pi f T_b)\right]$$

$$= \frac{-1}{j\pi f}\left[2\sin^2\left(\frac{\pi f T_b}{2}\right)\right]$$

$$= \frac{-1}{j\pi f}\left[2\sin\left(\frac{\pi f T_b}{2}\right)\sin\left(\frac{\pi f T_b}{2}\right)\right]$$

$$= \frac{-1}{j\pi f} \times \frac{T_b}{T_b}\left[2\sin\left(\frac{\pi f T_b}{2}\right)\sin\left(\frac{\pi f T_b}{2}\right)\right]$$

$$X(f) = jT_b \operatorname{sinc}\left(\frac{\pi f T_b}{2}\right)\sin\left(\frac{\pi f T_b}{2}\right)$$

Step 2: Autocorrelation $R_A(n) = E[A_L \cdot A_{L-n}]$

i) $n = 0$

$$R_A(0) = E[A_L A_L]$$

Bit	A_L	A_L	A_L^2	probability
0	$-A$	$-A$	A^2	$1/2$
1	$+A$	$+A$	A^2	$1/2$

$$R_A(0) = \frac{A^2}{2} + \frac{A^2}{2} = A^2$$

ii) $n \neq 0$

$$R_A(n) = E[A_L \; A_{L-n}]$$

Bit	A_L	A_{L-n}	$A_L \cdot A_{L-n}$	Probability
0 0	$-A$	$-A$	A^2	$1/4$
0 1	$-A$	$+A$	$-A^2$	$1/4$
1 0	$+A$	$-A$	$-A^2$	$1/4$
1 1	$+A$	$+A$	A^2	$1/4$

$$R_A(n) = \frac{A^2}{4} - \frac{A^2}{4} - \frac{A^2}{4} + \frac{A^2}{4} = 0$$

$$R_A(n) = \begin{cases} A^2, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Step 3: Find PSD.

$$P(f) = \frac{1}{T_b} |X(f)|^2 \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

$$= \frac{1}{T_b} \left| j T_b \operatorname{sinc}\left(\frac{\pi f T_b}{2}\right) \sin\left(\frac{\pi f T_b}{2}\right) \right|^2 \begin{cases} A^2 \cdot 1, & n \neq 0 \\ 0, & n = 0 \end{cases}$$

$$P(f) = \frac{1}{T_b} T_b^2 \operatorname{sinc}^2\left(\frac{\pi f T_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{2}\right) A^2$$

$$P(f) = A^2 T_b \operatorname{sinc}^2\left(\frac{\pi f T_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{2}\right)$$

This is the PSD of manchester polar format.

Unipolar NRZ:

$$P(f) = \frac{A^2}{4} T_b \operatorname{sinc}^2(\pi f T_b) + \frac{A^2}{4} \delta(f)$$

Polar NRZ:

$$P(f) = T_b \operatorname{sinc}^2(\pi f T_b) A^2$$

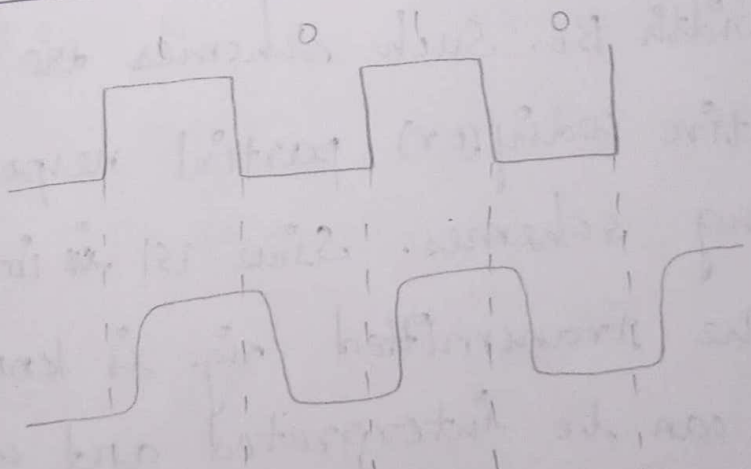
Bipolar NRZ:

$$P(f) = T_b A^2 \operatorname{sinc}^2(\pi f T_b) \sin^2(\pi f T_b)$$

Manchester polar format:

$$P(f) = A^2 T_b \operatorname{sinc}^2\left(\frac{\pi f T_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{2}\right)$$

ISI - Inter Symbol Interference:



Spreading of a pulse beyond its interval will cause it to interfere with neighbouring pulses resulting in intersymbol interference.

In the absence of ISI and channel noise, the transmitted signal can be decoded correctly in the receiver. But due to the occurrence of ISI, error will be introduced at decision making device in the receiver side.

Correlative Coding: done at the channel.

It is a process of increasing channel capacity by introducing ISI manually ($B_o \rightarrow 2B_o$) (changing 2 level signal to 3 level signal).

$\Rightarrow$ By adding ISI to transmitted signal in a controlled manner, it is possible to achieve the bit rate of $2B_o$ bps in a channel of bandwidth B_o. Such schemes are called correlative coding (or) partial response signaling schemes. Since ISI is introduced into the transmitted sig. is known, its effect can be interpreted and compensated at the receiver.

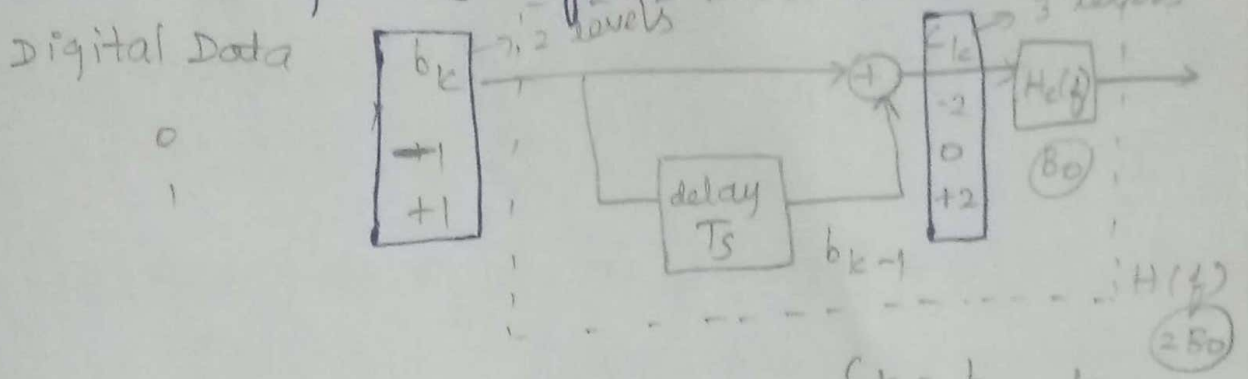
Methods of correlative coding:

1. Duobinary correlative coding
2. Modified correlative coding

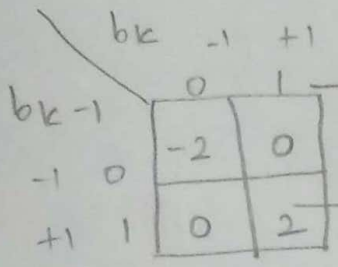
Duobinary correlative coding:

⇒ Encoder ⇒ Decoder ⇒ Freq. response.

The basic idea in duobinary method is doubling the transmission capacity of the binary system.



$c_k = b_k + b_{k-1}$



a_k = 0 or 1 (I/p bits)

b_k = +1, a_k=1 Bits are converted to polar form.

-1, a_k=0

Duobinary encoder produces $c_k = b_k + b_{k-1}$

o/p can have 3 levels c_k = -2, 0, +2

Input a_k	Polar b_k	Previous b_{k-1}	Duobinary o/p.
1	1	1	2
1	1	-1	0
0	-1	1	0
0	-1	-1	-2